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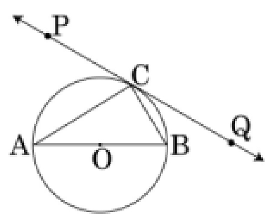
MATHEMATICS

SECTION A

1. In Fig., PQ is a tangent at point C to a circle with centre O. If AB is a diameter and $\angle CAB = 30°$, find $\angle PCA$.



2. For what value of k will $k+9$, $2k-1$ and $2k+7$ be the consecutive terms of an A.P.?

3. A ladder, leaning against a wall, makes an angle of $60°$ with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder.

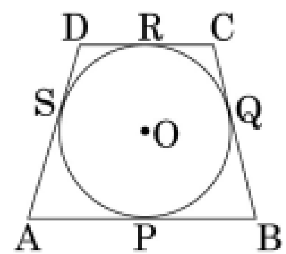
4. A card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen.

SECTION B

5. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find the value of k.

6. Let P and Q be the points of trisection of the line segment joining the points $A(2, -2)$ and $B(-7, 4)$ such that P is nearer to A. Find the coordinates of P and Q.

7. In Fig., a quadrilateral $ABCD$ is drawn to circumscribe a circle with centre O such that the sides AB, BC, CD and DA touch the circle at the points P, Q, R and S respectively. Prove that $AB + CD = BC + DA$.

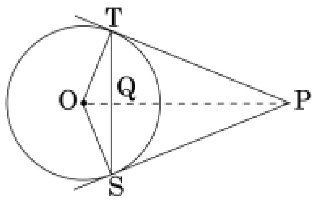


8. Prove that the points $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle.

9. The 4th term of an A.P. is zero. Prove that the 25th term of the A.P. is three times its 11th term.

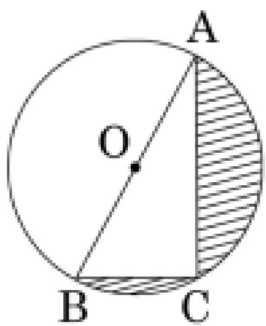
10. In Fig., from an external point P, two tangents PT and PS are drawn to a circle with centre O and radius r. If $OP = 2r$, show that

$$\angle OTS = \angle OST = 30°.$$



SECTION C

11. In Fig., O is the centre of a circle such that diameter $AB = 13$ cm and $AC = 12$ cm. BC is joined. Find the area of the shaded region. (Take $\pi = 3.14$)



12. In Fig., a tent is in the shape of a cylinder surmounted by a conical top of the same diameter. If the height and diameter of the cylindrical part are 2.1 m and 3 m respectively and the slant height of the conical part is 2.8 m, find the cost of canvas needed to make the tent if the canvas is available at the rate of Rs. 500 per sq. metre. (Use $\pi = \dfrac{22}{7}$)